
Module 1: Introduction to Big Data (9 hrs)

1. What is Big Data? How does it differ from conventional data?
 2. Explain the architecture of Big Data with a suitable diagram.
 3. List and describe the major Big Data platforms available today.
 4. What are the different types of data in Big Data? Provide examples.
 5. What are the key analytic processes in Big Data? Name some common tools used.
 6. Describe the 5 V's of Big Data with relevant examples.
 7. Explain different Big Data analytical methods and their applications.
 8. What is Intelligent Data Analysis? How does it differ from traditional analysis?
 9. Describe the stages in the Big Data Analytics Life Cycle.
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Module 2: Introduction to Stream Computing (8 hrs)

1. What is stream computing? How does it differ from batch processing?
 2. Explain the architecture of streaming data with an example.
 3. What is a stream data model? How is it different from traditional databases?
 4. Why is sampling important in stream processing? Describe different sampling techniques.
 5. What is a Bloom filter? How is it used for filtering streams?
 6. Explain the Flajolet-Martin algorithm and its application in stream data.
 7. How can moments be estimated in data streams?
 8. Describe the DGIM algorithm and its significance in stream computing.
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Module 3: Hadoop Distributed File System (13 hrs)

1. What is the history of Hadoop? Why was it developed?
 2. Describe the core components of the Hadoop ecosystem.
 3. Explain the architecture of HDFS with a labeled diagram.
 4. How does HDFS store and manage files? What is its design philosophy?
 5. Explain the roles of Blocks, Namenodes, and Datanodes in HDFS.
 6. What are the basic file system operations in Hadoop?
 7. List and describe different Hadoop-specific file types.
 8. Explain the process of reading a file in HDFS.
 9. How does HDFS write a file? Describe the key steps involved.
 10. What is an execution pipeline in Hadoop? How does it optimize data processing?
 11. Explain the role of runtime coordination and task management in MapReduce.
 12. How does MapReduce enable parallel processing? Give an example.
 13. Describe the Road Enrichment example in Hadoop.
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Module 4: Pig, Hive, and HBase (7 hrs)

1. What is Apache Pig? How does it simplify Big Data processing?
 2. Explain different execution modes of Pig with examples.
 3. Compare Pig with traditional databases. What are the advantages?
 4. What is Grunt in Pig? How does it help in processing data?
 5. Explain Pig Latin and its key features.
 6. What are user-defined functions in Pig? How are they used?
 7. Describe Hive and its architecture.
 8. What are the different services provided by Hive?
 9. Explain the role of Hive Metastore and how it compares with traditional databases.
 10. What is HiveQL? How does it simplify querying in Big Data?
 11. Explain different types of tables in Hive.
 12. What is HBase? How does it differ from RDBMS?
 13. Describe the basic concepts and architecture of HBase.
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Module 5: Introduction to R Programming (8 hrs)

1. What is R? Why is it popular for data analysis?
 2. Describe the key features of the R language.
 3. Explain different types of R Graphical User Interfaces.
 4. What are vectors in R? How are they created and used?
 5. How can matrices be created in R? Explain with an example.
 6. How do you apply functions to matrix rows and columns in R?
 7. What are lists in R? How are they different from vectors and matrices?
 8. Explain general list operations in R with examples.
 9. How can multiple variables be examined in R?
 10. What are data frames in R? How do they differ from matrices?
 11. Explain matrix-like operations in data frames with examples.
 12. How do you apply functions to data frames? Provide an example.
 13. How can files be read and written in R?
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